

1. List the user commands and redirect the output to /tmp/commands.

```
sudo ls /usr/bin/ >> /tmp/commands
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ sudo ls /usr/bin/ >> /tmp/commands  
[sudo] password for nagy:  
nagy@ZenBook-UX425EA:~$
```

2. Count the number of user commands

```
wc -l /tmp/commands
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ wc -l /tmp/commands  
5012 /tmp/commands  
nagy@ZenBook-UX425EA:~$
```

3. Get all the users names whose first character in their login is 'g'.

```
cat /etc/passwd | cut -f1 -d: | grep ^g
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ cat /etc/passwd | cut -f1 -d: | grep ^g  
games  
gnats  
geoclue  
gnome-initial-setup  
gdm  
nagy@ZenBook-UX425EA:~$
```

4. Get the logins name and full names (comment) of logins starts with "g".

```
cat /etc/passwd | cut -f1,5 -d: | grep ^g
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ cat /etc/passwd | cut -f15 -d: | grep ^g  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$
```

5. Save the output of the last command sorted by their full names in a file.

```
cat /etc/passwd | cut -f1 -d: | grep ^g | sort -t : -k1
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ cat /etc/passwd | cut -f1 -d: | grep ^g | sort -t : -k1  
games  
gdm  
geoclue  
gnats  
gnome-initial-setup  
nagy@ZenBook-UX425EA:~$
```

7. Display the number of users who is logged now to the system.

```
w | wc -l
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ w  
13:47:51 up 17:29, 1 user, load average: 0.51, 0.46, 0.48  
USER      TTY      FROM          LOGIN@   IDLE   JCPU   PCPU   WHAT  
nagy      tty2     tty2          Mon20    17:29m  0.01s  0.00s  /usr/libexec/gdm-  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$
```

8. Display lines 7 to line 10 of /etc/passwd file

```
cat /etc/passwd | head -10 | tail -7
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ cat /etc/passwd | head -10 | tail -7  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$
```

9. What happens if you execute:

- `cat filename1 | cat filename2`

it executes the first cmd "cat" then pass the file output as an input to the next cat which is useless, because it have already input file

- `ls | rm`

it ignores it because it needs the bachtec ``

- `ls /etc/passwd | wc -l 1`

10. Issue the command sleep 100.

it waits for 100 seconds so the cpu will be holded

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ sleep 100
```

11. Stop the last command.

`sudo kill -STOP 119365`

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ sudo kill -STOP 119365  
[sudo] password for nagy:  
  
[1]+  Stopped                  sleep 100  
nagy@ZenBook-UX425EA:~$
```

12.Resume the last command in the background

```
sudo kill -CONT 119365
```

```
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$ sudo kill -CONT 119365  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$  
nagy@ZenBook-UX425EA:~$
```

13.Issue the jobs command and see its output.

```
jobs
```

14.Send the sleep command to the foreground and send it again to the background.

```
nagy@ZenBook-UX425EA:~$ fg %1  
sleep 100  
^Z  
[1]+  Stopped                  sleep 100
```

15.Kill the sleep command.

```
sudo kill -9 120339
```

16.Display your processes only

```
ps -u $USER
```

17.Display all processes except yours

```
pgrep -v -u $USER
```

18.Use the pgrep command to list your processes only

```
pgrep -u $USER
```

19.Kill your processes only.

```
pkill -u $USER
```
